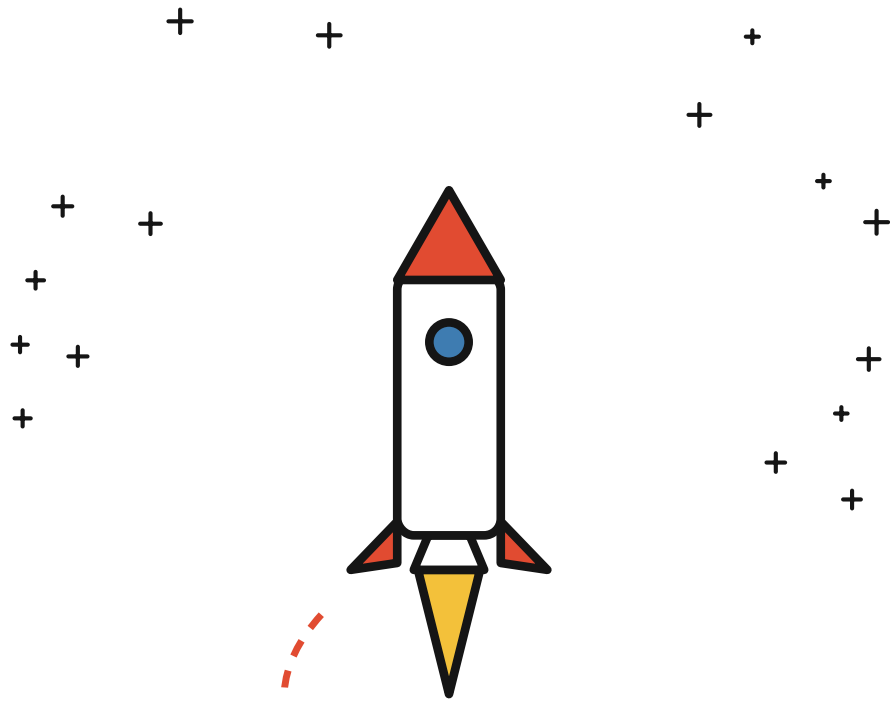


**Next:**

## Doing It Again

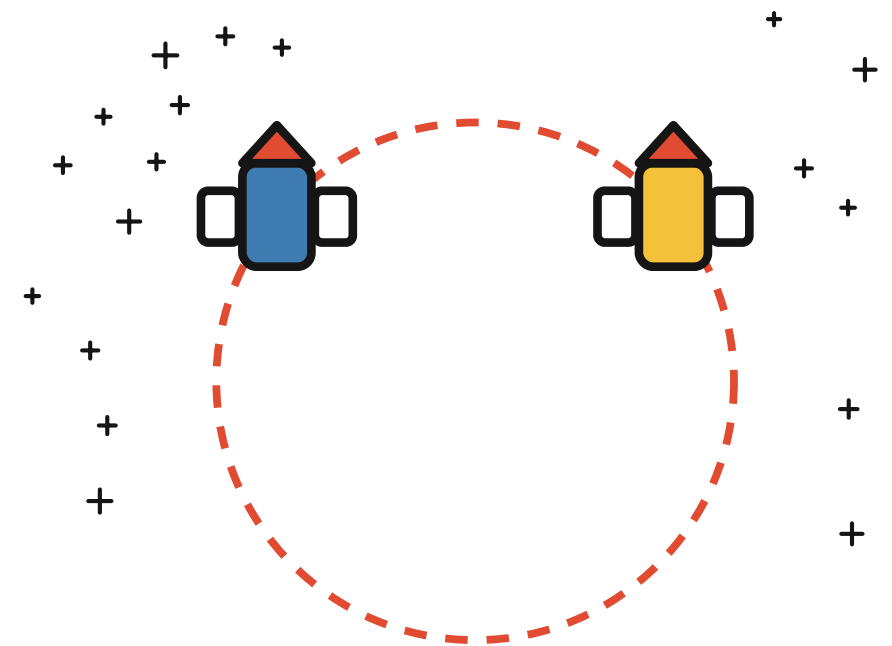
Book Eight · why we stopped throwing rockets away



Every rocket so far  
was used only once.

# FINDING EACH OTHER

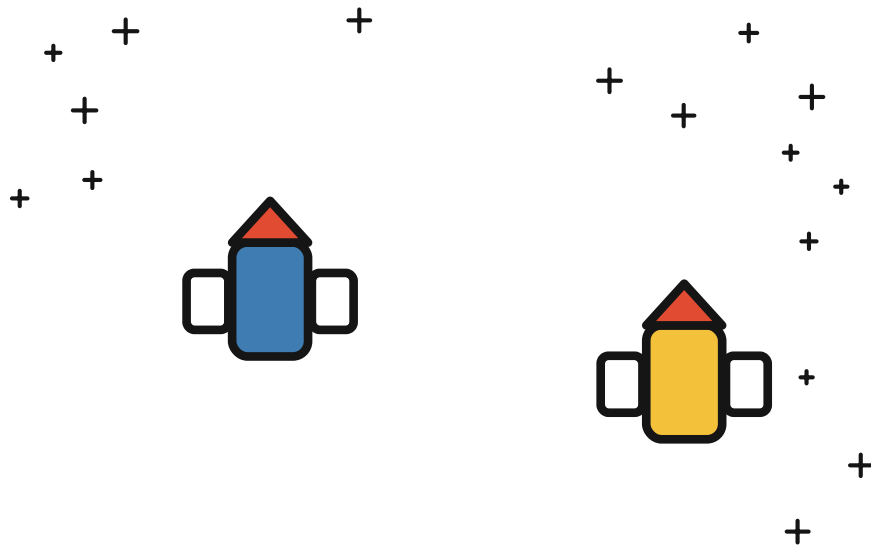
Book Seven · How two ships meet when both are falling



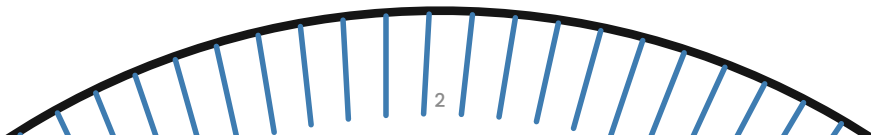
Print it. Read it together. Then race him round the table.

# Two ships up there.

Both falling. Both going very fast.



You want to get from one to the other.



# For the grown-up.

## The one idea

Orbital mechanics is not driving. Thrusting towards a target raises your orbit, which lengthens your period and drops you behind it. To catch something ahead of you, you burn retrograde, fall into a lower and quicker orbit, gain ground, then raise back up to meet it. Nearly every intuition from moving on a surface is inverted.

## Gemini

The entire programme existed to learn this. Gemini 6A and 7 flew in formation on 15 December 1965, holding station a few metres apart. Gemini 8, with Armstrong and Scott, achieved the first docking on 16 March 1966 — and then a stuck thruster spun the joined craft up to roughly one revolution per second, forcing an emergency undock and early return.

## The Moon plan

Lunar-orbit rendezvous was championed by John Houbolt at Langley, against considerable institutional resistance, and selected in July 1962. Direct ascent would have required the never-built Nova rocket. LOR wins because the lander never carries the Earth-return propellant down to the surface and back up — the mass saving cascades all the way back to the pad.

## What it cost in nerve

The choice meant that if the ascent stage failed to reach orbit, or the two craft failed to find each other, there was no rescue possible. Rendezvous went from a research problem to the load-bearing element of the whole endeavour.

# Try this tonight.

## 1 Race him round the table.

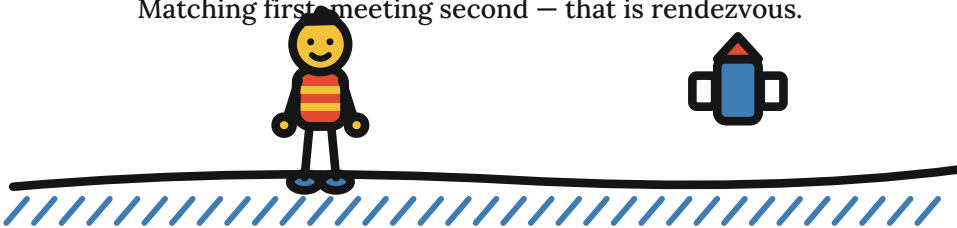
You take the outside, he takes the inside, and start level. He wins every time without going faster. That is exactly why a lower orbit laps a higher one.

## 2 Swing a ball on a string.

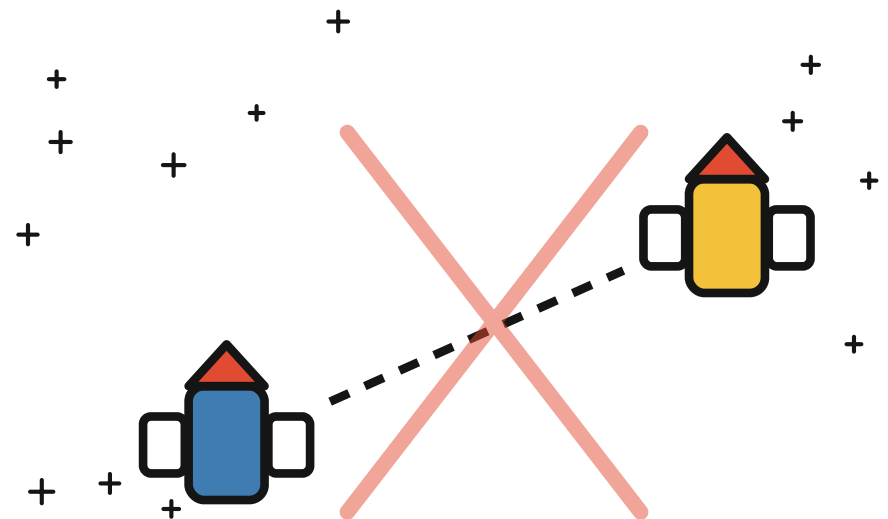
Short string, quick circles. Let out more string and the circles slow right down. Bigger loop, slower lap – the same rule that makes speeding up put you behind.

## 3 Try to hand something over while walking.

Walk side by side and pass a toy back and forth. Easy. Now try it while one of you speeds up. Matching first meeting second – that is rendezvous.

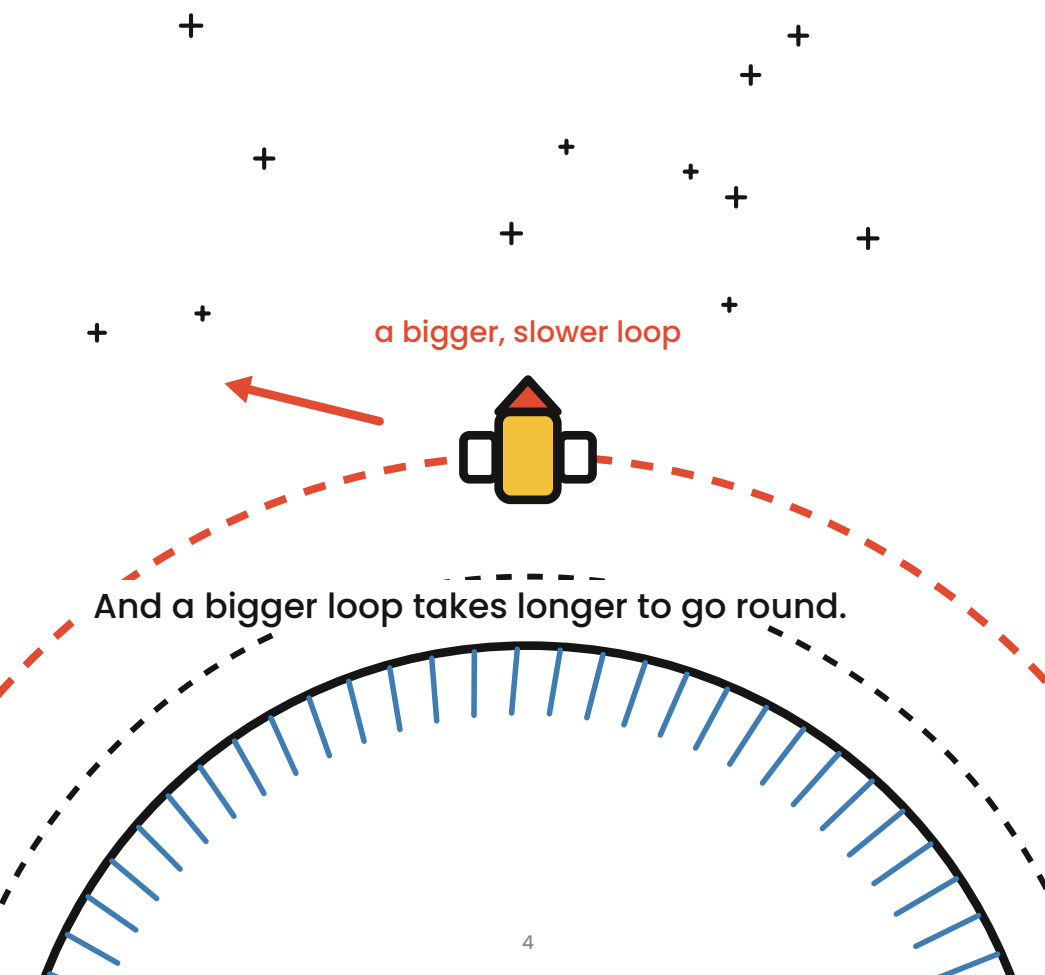


# So point at it and go, yes?



No. Up here, that does not work at all.

# Speed up, and you climb higher.



## Three ways to reach the Moon.

Finding Each Other · figure 2 · the choice made in 1962

### Direct ascent

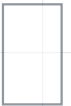
one enormous rocket  
lands the whole ship  
needs a launcher far bigger  
than any ever built



rocket needed

### Earth-orbit rendezvous

two launches, joined  
in orbit around Earth  
still lands the whole ship  
on the Moon



rocket needed

### Lunar-orbit rendezvous

leave the big ship circling  
the Moon  
a small lander goes down  
and comes back up



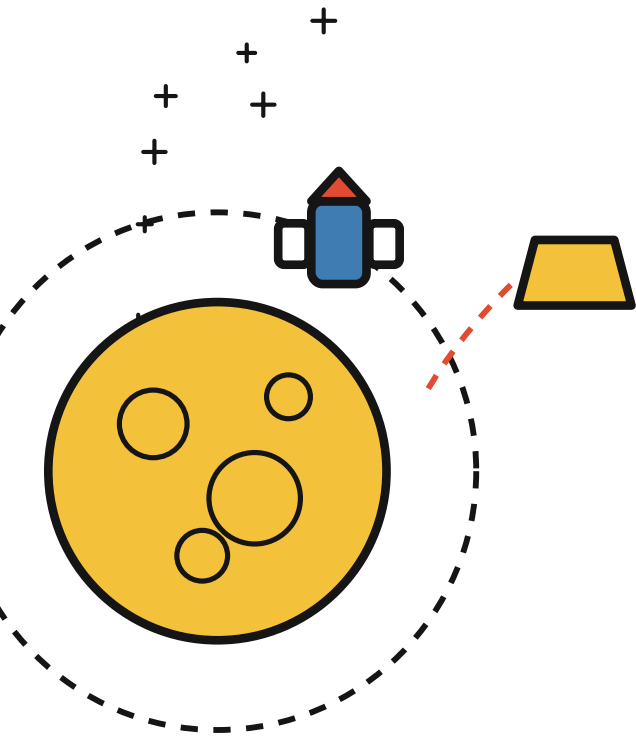
rocket needed

chosen

Lunar-orbit rendezvous won because it is by far the lightest —  
you never haul the return fuel down to the surface and back up.

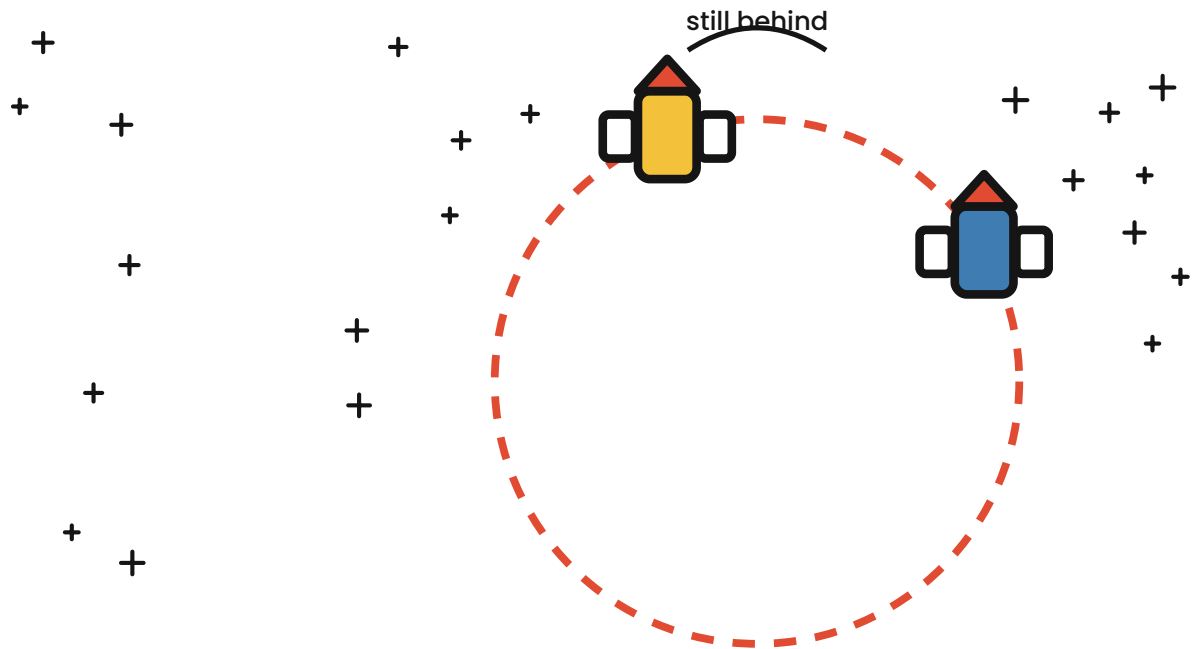
It also meant everything depended on two ships meeting, 380,000 km from help.

**But then you must  
find it again.**



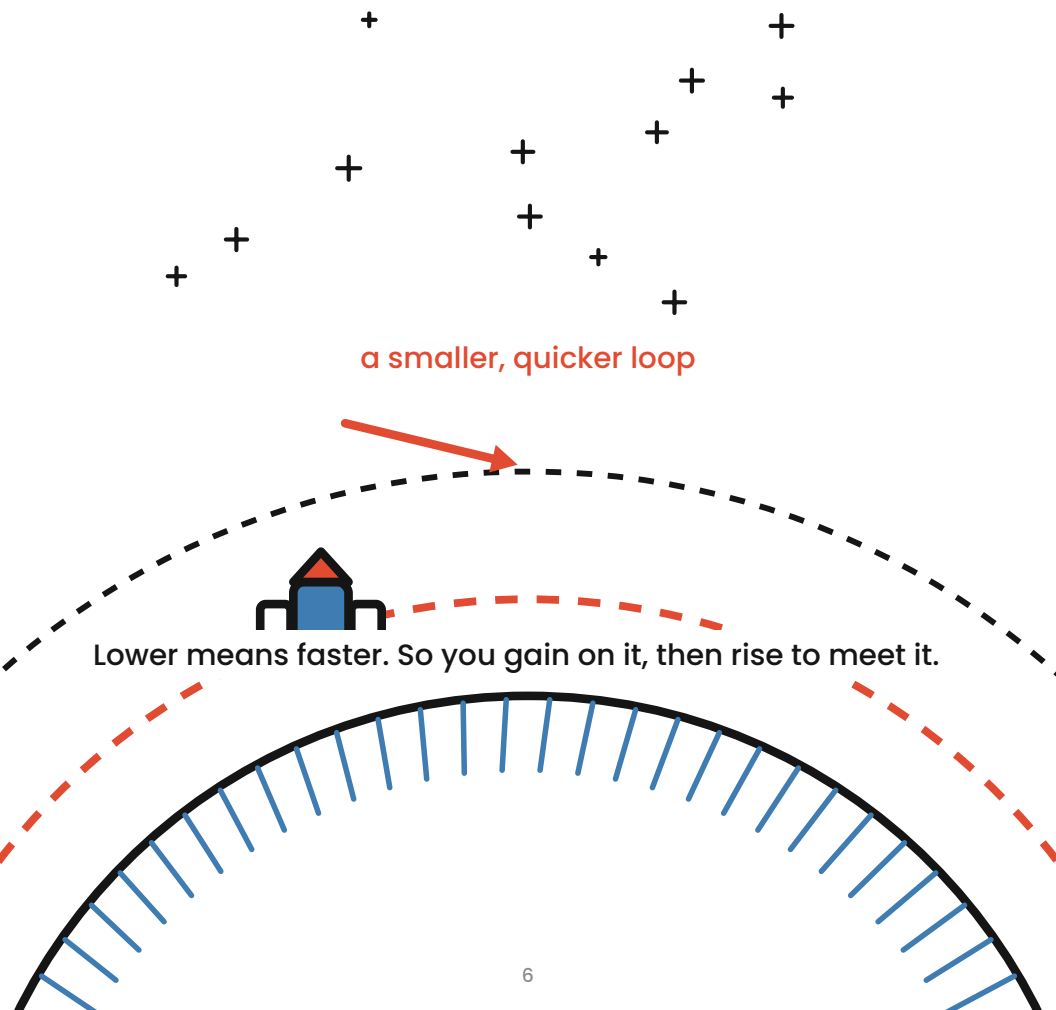
Two tiny things, above another world,  
**with no one to help you.**

**So going faster  
makes you late.**



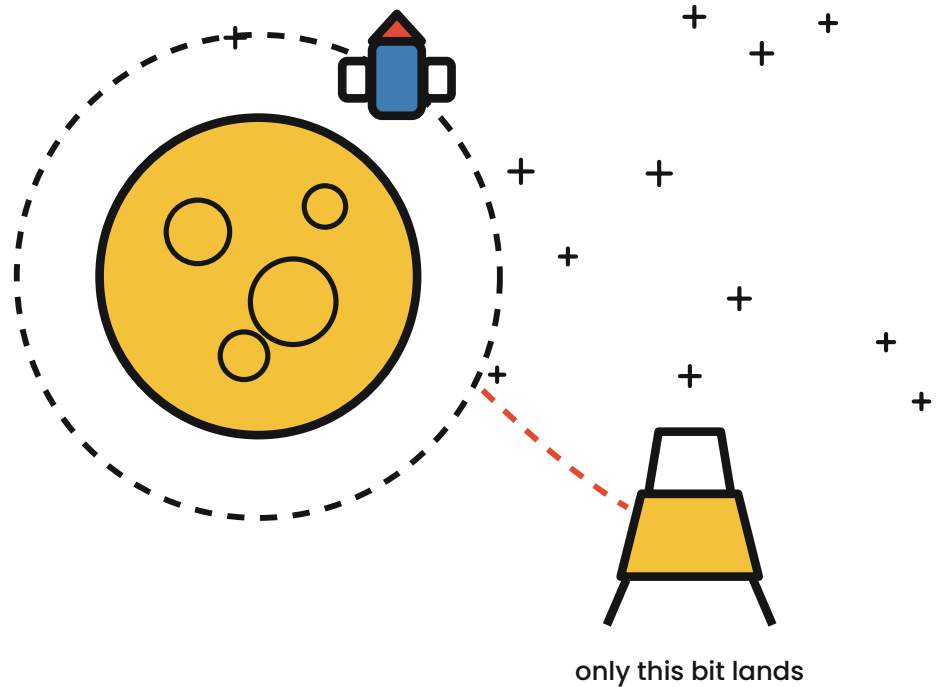
You went faster, and fell further behind.  
**That is the strangest thing in space.**

To catch up,  
**drop down lower.**



Or do you?

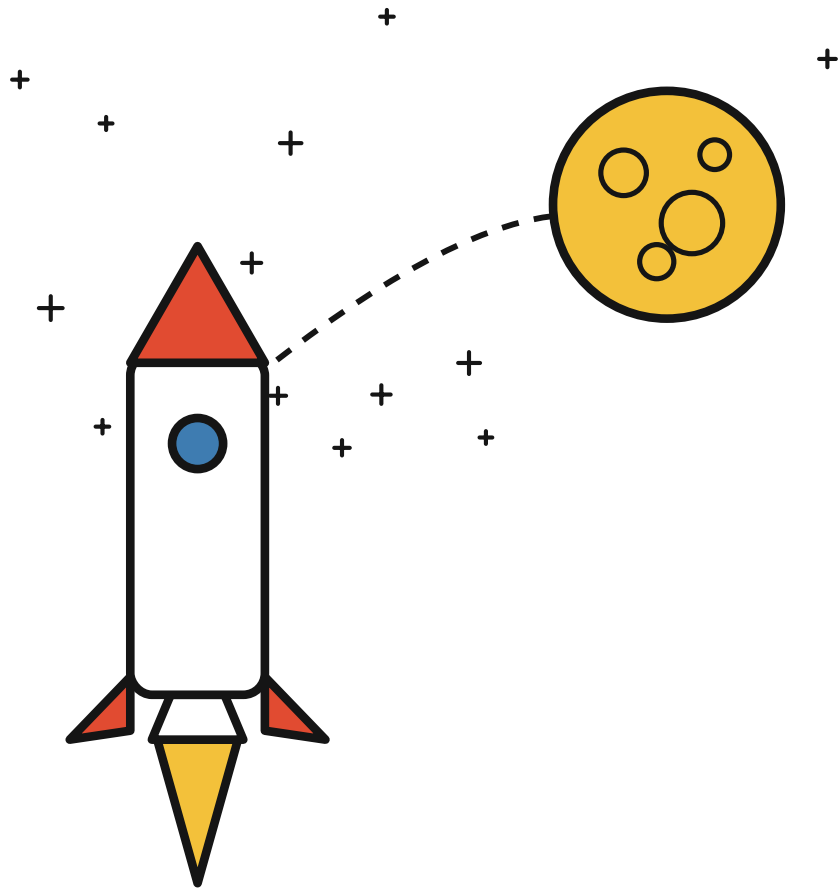
Here was the clever idea.



Leave most of the ship in a loop, and send down a little one.

# Why bother?

Because of the Moon.

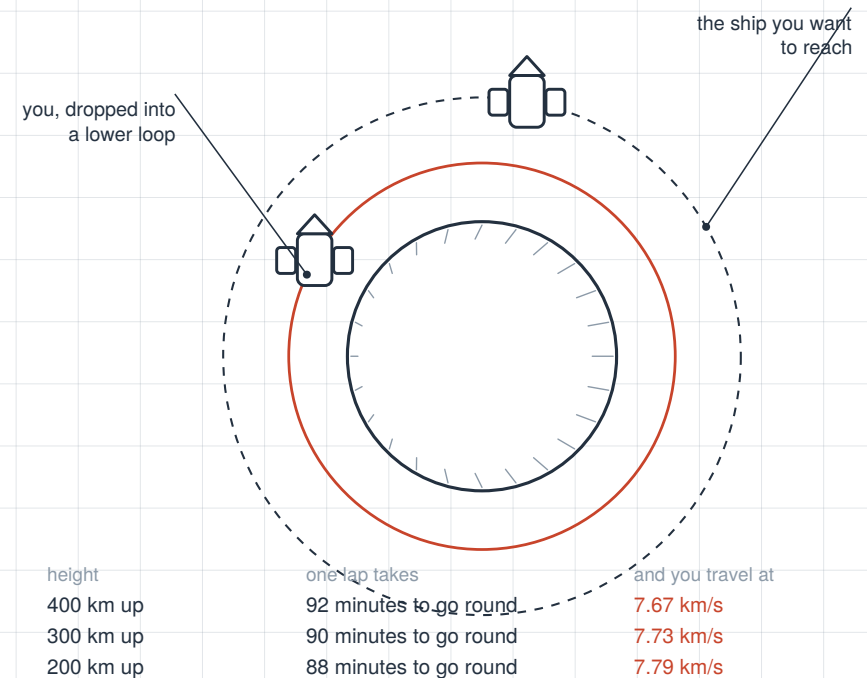


To get there, you have to take everything with you.

10

# The chase.

Finding Each Other · figure 1 · why lower is quicker

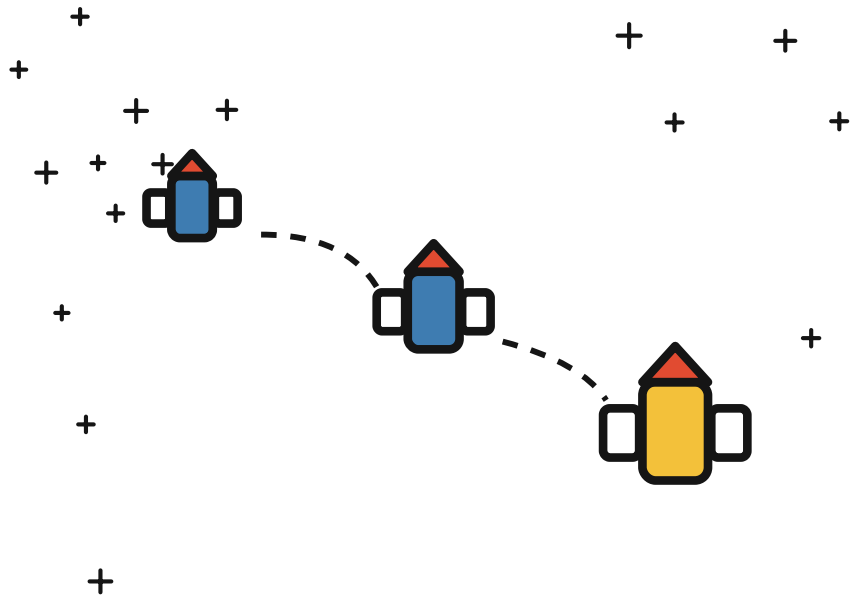


Lower orbits are both faster and shorter, so they lap higher ones.  
To catch something ahead of you, you fire backwards and drop.

Every instinct from driving a car is wrong here.

7

**They practised  
again and again.**



Closer. Then closer. Then closer still.  
**It took them ten tries to get good at it.**

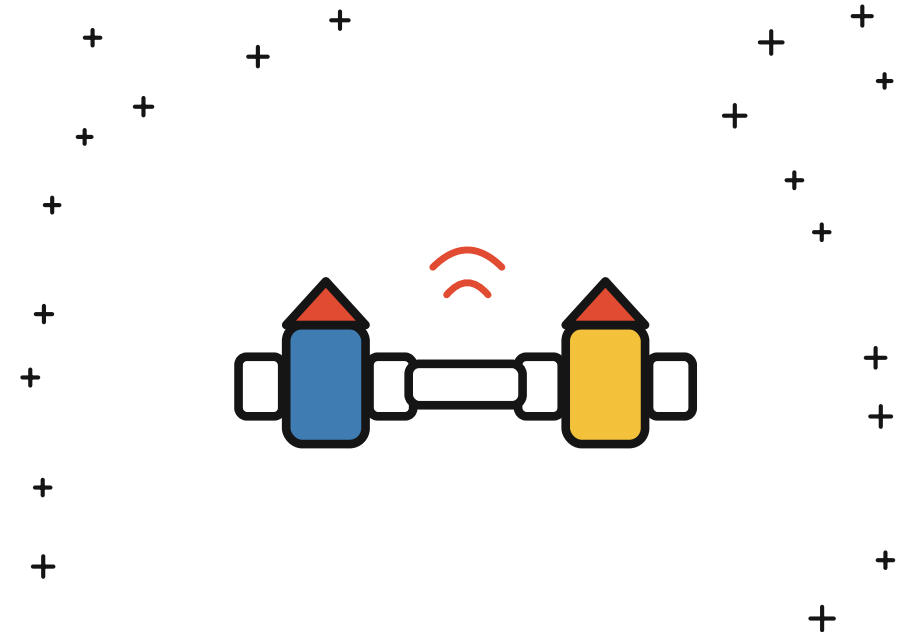
8

**Then two ships  
touched.**

pierce here

pierce here

pierce here



For the first time, in 1966,  
**two spacecraft were joined together.**

9

fold along this line